

CHAPTER-1: INTRODUCTION

1.1 Overview

In recent years, the discovery of career opportunities such as internships, full-time roles, and hackathons has become highly fragmented. Students and early-career professionals often struggle to monitor multiple portals (e.g., LinkedIn, Devfolio, Internshala, Unstop) simultaneously. OpportunityIQ addresses this by acting as a centralized intelligence system. It automatically scrapes, aggregates, and processes opportunities, delivering them to users via an intuitive web dashboard and Telegram bot.

1.2 Generative AI in Opportunity Matching

The core innovation of OpportunityIQ is the shift from traditional keyword-based filtering to semantic opportunity matching powered by Large Language Models (LLMs). By understanding the context and adjacent skills associated with a candidate's profile, the system provides accurate recommendation scores, vastly improving discovery for non-standard job descriptions and resolving the cold-start problem in recruitment platforms.

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CHAPTER-2: LITERATURE SURVEY

A literature review is a survey of scholarly sources on a specific topic. It provides an overview of current knowledge, allowing you to identify relevant theories, methods, and gaps in the existing research.

2.1 Critical Analysis of the literature

The integration of Artificial Intelligence in recruitment and career recommendation has seen a significant shift from rule-based engines to deep learning models between 2019 and 2022. A critical analysis of the literature reveals several dominant themes in the domain:

1. Transition to Semantic Skill Matching

Traditional job recommendation relied heavily on strict lexical matching, leading to poor discovery rates. Research by Smith et al. [1] in IEEE Transactions and Zhao et al. [7] in Elsevier highlighted the limitations of string matching, proposing knowledge graph-based context-aware recommendations. Furthermore, studies [2, 12, 19] demonstrated that transformer-based models and semantic ontologies significantly outperform traditional models by understanding domain adjacency. This aligns directly with OpportunityIQ's use of Generative AI for taxonomy expansion.

2. Data Extraction and Aggregation from Unstructured Sources

The literature heavily focuses on gathering opportunities from disparate sources. Works by Kumar et al. [6] and Li et al. [15] emphasized the effectiveness of NLP and BERT models for extracting structured data from unstructured web job posts. Additionally, scalable architectures for multi-platform job aggregation were discussed in [18], proving the necessity of robust data pipelines similar to OpportunityIQ's node-cron based ingestion engine.

3. Addressing the Cold-Start Problem and Personalization

A recurring challenge in literature is the "cold-start" problem where new users lack sufficient profile data. ACM publications [4, 9] introduced multi-source data fusion and skill taxonomy expansion as viable solutions. Research by Chen et al. [16] in Elsevier focused on personalized career trajectories, showing that inferring skills from minimal input greatly enhances recommendation relevance. OpportunityIQ solves this via Synthetic Taxonomy Expansion using LLMs.

4. Bias, Fairness, and Real-time Communication

Recent literature also stresses the ethical dimensions and user experience in automated recruitment. IEEE studies [14] critically assessed fairness in AI-driven systems, while works like [10, 20] explored the implementation of message queues and chatbot integrations for real-time candidate notifications. OpportunityIQ incorporates these principles by ensuring transparent LLM-based reasoning and utilizing a Telegram Bot for real-time alerts.

2.2 Implication and conclusion

The literature establishes a clear transition from basic keyword aggregators to intelligent, semantic-driven matching systems. While existing research heavily documents the use of BERT and Graph Networks for skill matching, there remains a notable gap in open-source, multi-platform aggregators that leverage zero-shot capabilities of modern LLMs for real-time taxonomy expansion. The implications suggest that an integrated platform—combining scalable web scraping with dynamic GenAI-driven skill evaluation—can significantly outperform isolated scraping tools. Therefore, OpportunityIQ is proposed as a novel synthesis of these academic concepts, providing a seamless bridge between fragmented job boards and intelligent matching.

2.3 Problem Statement

The current landscape of career opportunity discovery is highly fragmented across dozens of platforms (LinkedIn, Internshala, Devfolio, etc.), requiring students to manually track and filter opportunities. Furthermore, existing alert systems rely on strict, keyword-based lexical matching, leading to high false-negative rates when job posters use alternative terminology. There is a critical need for an automated, centralized intelligence system capable of aggregating multi-source data and utilizing semantic AI understanding to accurately match opportunities with candidate profiles.

2.4 Objectives

1. To design and implement a scalable, automated ingestion engine that aggregates opportunities across multiple platforms in real-time.
2. To develop a GenAI-powered semantic matching system capable of context-aware skill evaluation and synthetic taxonomy expansion to eliminate the cold-start problem.
3. To deploy a real-time notification mechanism and an interactive web dashboard for seamless user profile management and personalized discovery.

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CHAPTER-3: REQUIREMENT ANALYSIS

3.1 Functional Requirements

- **Multi-Source Ingestion:** The system must scrape data from predefined platforms (Devfolio, Internshala, LinkedIn, Greenhouse) periodically.
- **AI-Powered Evaluation:** The system must parse job descriptions using an LLM to generate a `relevancePct` and `matchedKeywords` based on the user's profile.
- **Real-Time Notifications:** A Telegram Bot must allow users to subscribe to domains and receive real-time alerts for highly matched roles.
- **Web Dashboard Management:** Users must be able to log in, update their skill sets, and view their personalized opportunity feed.

3.2 Non-Functional Requirements

- **Scalability:** The architecture must handle a high volume of scraped jobs daily without degrading notification latency.
- **Reliability:** Background scraping tasks must be fault-tolerant and gracefully handle layout changes on target websites.
- **Security:** User email authentications and profile data must be securely stored using Supabase RLS policies.
- **Performance:** AI evaluations should ideally be processed asynchronously to ensure non-blocking UI experiences.

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CHAPTER-4: SYSTEM DESIGN

The system architecture of OpportunityIQ is divided into three primary tiers:

1. **Data Ingestion & Processing Layer (Backend):** Built with Node.js, Express, and Cheerio, running automated CRON jobs to scrape platforms. The data is structured, sanitized, and stored centrally.
2. **AI Intelligence Engine:** Leveraging the Groq/Gemini API, this module receives unstructured job data and evaluates it against user profiles. It performs Synthetic Data Expansion and outputs a strict JSON containing the contextual match score.
3. **Presentation & Notification Layer:**
 - **Web Application:** A frontend built with HTML/JS and Tailwind CSS providing a sleek, dark-themed dashboard for viewing matched jobs.
 - **Telegram Bot Integration:** Uses Telegraf.js to act as a conversational interface for mobile-first users, sending formatted MarkdownV2 alerts.
4. **Database:** Supabase (PostgreSQL) is used as the central repository for users, opportunities, and matches, equipped with Edge Functions for triggering real-time webhooks.

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CHAPTER-5: METHODOLOGY

The development methodology for OpportunityIQ follows an Agile, modular approach:

1. **Scraping Module Development:** Initial phase focused on building custom adapters for different DOM structures of target job boards.
2. **Database Schema Design:** Establishing relational mapping between opportunities and users in Supabase, optimizing for read-heavy operations on the dashboard.
3. **LLM Prompt Engineering:** Iterative testing of prompts using zero-shot classification to ensure the GenAI model accurately acts as a technical recruiter and outputs consistent JSON.
4. **Integration of Bot & Web Dashboard:** Hooking up the Express backend to serve REST APIs consumed by the Web dashboard, and integrating Telegram long-polling for bot commands.
5. **Deployment:** The Node.js application is containerized and deployed on Render with persistent background workers, while the web dashboard is hosted on Vercel.

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CHAPTER-6: CONCLUSION AND FUTURE WORK

Conclusion:

OpportunityIQ successfully bridges the gap between fragmented career platforms and modern job-seekers. By transitioning from rigid lexical queries to dynamic, LLM-powered semantic matching, the platform drastically improves the relevance of recommended jobs. The successful integration of multi-source scraping, a Telegram notification bot, and a responsive web dashboard proves the feasibility and utility of an AI-driven career intelligence ecosystem.

Future Work:

Future enhancements will focus on replacing periodic scraping with real-time webhooks where possible to reduce latency. Additionally, the integration of fine-tuned, smaller language models (such as LLaMA-3) directly on edge devices or private servers could reduce API costs and improve privacy. Expanding the taxonomy to include automated resume parsing from uploaded PDFs will further streamline user onboarding.

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